

Liu Restrepo Sanabria

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TECHNICAL SKILLS

- **HDL & FPGA:** Verilog, VHDL, Vivado, Vitis; PLL configuration, FSM design, Xilinx Artix-7, ARM Cortex-M.
- **Embedded Systems:** Embedded C, Assembly (8051), ESP32, UART, I2C, SPI, RTOS.
- **Languages and Tools:** C/C++, Python, MATLAB; KiCad, LTSpice, AutoCAD, Keil MDK; Linux, Bash, Git.

EDUCATION

Trinity College

Aug 2023 – Expected May 2027

Bachelor of Science, Electrical and Computer Engineering and Computer Science

Hartford, CT, USA

- **GPA:** 4.13/4.00
- **Scholarships & Honours:** Davis, Dean's, and Moran's Scholar; Faculty Honours (x4).
- **Coursework:** Embedded Systems, Signal Processing, High-Performance Computing, Digital Communications.

University of Edinburgh

Aug 2025 – December 2025

Electrical and Electronics Engineering and Informatics

Edinburgh, UK

- **Coursework:** Embedded Programming, Computer Systems, Analogue Circuits.

EXPERIENCE

Teaching Assistant – Embedded Systems Design

Jan 2026 – Current

Department of Engineering at Trinity College

Hartford, CT, USA

- Support lab instruction for 8 students in ENGR-323, covering microcontroller firmware, hardware interfacing, and real-time system design.
- Guide students in debugging 8051 Assembly and C firmware, focusing on interrupt-driven architectures, timers, and serial protocols (I2C, UART.)

Undergraduate Research Assistant

Apr 2025 – Current

Department of Engineering at Trinity College

Hartford, CT, USA

- Developed nonlinear signal processing algorithms in MATLAB and FORTRAN (including a normalised Grassberger–Procaccia estimator) to extract fractal dimensions from noisy multi-channel hippocampal EEG recordings.
- First-authored a peer-reviewed IEEE conference paper published in IEEE Xplore (CISP-BMEI 2025, Qingdao, China).

Undergraduate Research Assistant

May 2025 – July 2025

Physics Department at Trinity College

Hartford, CT, USA

- Redesigned and optimized a 4-channel photon coincidence counting system on an Alchitry Au Xilinx Artix-7 FPGA (Verilog HDL,) targeting throughput and timing stability constraints imposed by high input event rates.
- Reduced time resolution by 67% and increased stable counting throughput by 175% by overclocking from 100 MHz to 300 MHz via PLL configuration, critical path analysis and FSM restructuring.
- Designed and implemented a UART interface and real-time PC GUI for low-latency FPGA data streaming and automated event logging, significantly improving system observability and debug efficiency.

PROJECTS

Computer Cluster for HPC applications

Mar 2026

High Performance Computing Project

Trinity College

- Built a 4-node HPC cluster from commodity hardware, configuring Linux networking, MPI parallelisation in C, and HPL benchmarking; achieved 242 GFLOPS peak performance across the distributed system.

Heartbeat Monitoring System

Apr 2025

Embedded Systems Design Project

Trinity College

- Designed an 8051-based heart rate monitor in Assembly: op-amp signal conditioning, edge-triggered interrupt pulse detection, scaled 8-bit BPM arithmetic (± 1 BPM, 50–120 BPM), and TDM 7-segment display at 4.3% CPU utilisation.